

CSC 306: Human Computer Interaction Design

Topic: Human Cognitive and Physical
Ergonomics

HCI

- Human-computer interaction (commonly referred to as **HCI**) researches the design and use of computer technology, focused on the interfaces between people (**users**) and **computers**. Researchers in the field of HCI both observe the ways in which humans interact with computers and design technologies that let humans interact with computers in novel ways.
- We have key words like **User** (Human), **Interaction** and **Computer**

User

- By "**user**", we may mean an individual user, a group of users working together. An appreciation of the way people's sensory systems (sight, hearing, touch) relay information is vital. Also, different users form different conceptions or mental models about their interactions and have different ways of learning and keeping knowledge and. In addition, cultural and national differences play a part.

Computer

- When we talk about the **computer**, we're referring to any technology ranging from desktop computers, to large scale computer systems. For example, if we were discussing the design of a Website, then the Website itself would be referred to as "the computer". Devices such as mobile phones or VCRs can also be considered to be —computers.

Interaction

- There are obvious differences between humans and machines. In spite of these, HCI attempts to ensure that they both get on with each other and interact successfully. In order to achieve a usable system, you need to apply what you know about humans and computers, and consult with likely users throughout the design process. In real systems, the schedule and the budget are important, and it is vital to find a balance between what would be ideal for the users and what is feasible in reality.

The Goals of HCI

- The goals of HCI are to produce **usable** and **safe** systems, as well as **functional** systems. In order to produce computer systems with good usability, developers must attempt to: understand the factors that determine how people use technology, develop tools and techniques to enable building suitable systems, achieve efficient, effective, and safe interaction put people first.
- Underlying the whole theme of HCI is the belief that people using a computer system should come first. Their needs, capabilities and preferences for conducting various tasks should direct developers in the way that they design systems. People should not have to change the way that they use a system in order to fit in with it. Instead, the system should be designed to match their requirements.

Usability

- Usability is one of the key concepts in HCI. It is concerned with making systems easy to learn and use. A usable system is:
 - easy to learn
 - easy to remember how to use
 - effective to use
 - efficient to use
 - safe to use
 - enjoyable to use

Factors in HCI

- There are a large number of factors which should be considered in the analysis and design of a system using HCI principles. Many of these factors interact with each other, making the analysis even more complex. The main factors are listed in the table below:

Factors in HCI

- **Organisation Factors**

- Training, job design, politics, roles, work organisation
- Environmental Factors
- Noise, heating, lighting, ventilation
- Health and Safety Factors

- **The User**

- Cognitive processes and capabilities
- Motivation, enjoyment, satisfaction, personality, experience
- Comfort Factors
- Seating, equipment, layout.

Factors in HCI

- **User Interface**

Input devices, output devices, dialogue structures, use of colour, icons, commands, navigation, graphics, natural language, user support, multimedia,

- **Task Factors** : Easy, complex, novel, task allocation, monitoring, skills
- **Constraints** : Cost, timescales, budgets, staff, equipment, buildings
- **System Functionality**: Hardware, software, application
- **Productivity Factors** : Increase output, increase quality, decrease costs, decrease errors, increase innovation

Disciplines contributing to HCI

- The field of HCI covers a wide range of topics, and its development has relied on contributions from many disciplines. Some of the main disciplines which have contributed to HCI are:
- **Computer Science**
 - technology
 - software design, development & maintenance
 - User Interface Management Systems (UIMS) & User Interface Development Environments (UIDE)
 - prototyping tools
 - graphics

Disciplines contributing to HCI

- **Cognitive Psychology**
 - information processing
 - capabilities
 - limitations
 - cooperative working
 - performance prediction
- **Social Psychology**
 - social & organizational structures
- **Ergonomics/Human Factors**
 - hardware design
 - display readability

Disciplines contributing to HCI

- **Linguistics**
 - natural language interfaces
- **Artificial Intelligence**
 - intelligent software Engineering & Design
 - graphic design
 - engineering principles

HUMAN COGNITIVE AND PHYSICAL ERGONOMICS

- **Ergonomics** (also known as Human Factors) is the scientific discipline concerned with understanding the interactions between humans and other elements of a system. In HCI, the golden rule of ergonomics is: *Design the system to fit the human, rather than forcing the human to alter their body or mind to fit the system.*
- In HCI, human beings are the bottleneck. Technology is incredibly fast, but human bodies get tired, and human brains get overwhelmed.
- If we ignore ergonomics, we will build systems that cause physical pain, mental exhaustion, catastrophic workplace errors, and product abandonment. To bridge this gap, ergonomics is split into two massive pillars: **Physical Ergonomics** (designing for the body) and **Cognitive Ergonomics** (designing for the mind).

1. Cognitive Ergonomics

- **Cognitive ergonomics** is all about how software accommodates human mental processes. It looks at our psychological limitations—like memory, attention, perception, and decision-making—and ensures the interface doesn't cause mental burnout.
- In software design, ignoring cognitive ergonomics leads to user frustration, high error rates, and system abandonment caused by physical pain, mental exhaustion, and catastrophic workplace errors.
- When we design interfaces, we are essentially building a bridge between two processors: the silicon processor inside the computer and the biological processor inside the human skull.
- **Core Focus Areas:**
 - **Mental Workload:** Preventing "cognitive overload." If an interface displays too much data at once, the user's brain freezes.
 - **Memory Constraints:** Human short-term memory can only hold about 7 ± 2 chunks of information at a time (Miller's Law). Good design relies on recognition rather than recall (e.g., using a visible menu instead of making a user remember complex command lines).
 - **Decision Making:** The more choices you give a user, the longer it takes them to make a decision (Hick's Law). Cognitive ergonomics pushes for clean, minimalist designs that guide the user smoothly.

1. Cognitive Ergonomics

- To turn cognitive ergonomics into hard science, HCI utilizes mathematical laws to optimize user performance.
- **Hick's Law** dictates that the time it takes for a user to make a decision increases logarithmically with the number of options presented. It is expressed mathematically as:

$$T = a + b \log_2(n + 1)$$

Where:

- T is the total reaction time.
- n is the number of equally probable choices.
- a and b are empirical constants representing cognitive processing speed.

Design Implication: Do not overwhelm users with massive, flat menus. Instead, utilize progressive disclosure, hierarchical categorization, or multi-step wizards to drastically minimize choice selection times.

1. Cognitive Ergonomics

- **Fitts's Law** is a biomechanical and cognitive model predicting the time required to rapidly move to a target area based on the **distance** to the target and the **size** of the target. The standard Shannon formulation is:

$$MT = a + b \log_2 \left(\frac{D}{W} + 1 \right)$$

Where:

- MT is the Movement Time to acquire the target.
- D is the Distance (amplitude) to the center of the target.
- W is the Width of the target along the axis of motion.
- The term $\log_2 \left(\frac{D}{W} + 1 \right)$ is known as the Index of Difficulty (ID).

Design Implication: Crucial UI elements (like a “Submit” button) should be large (W) and placed close to the user's expected cursor or hand placement (D). Elements placed at the edge of screens or corners require less precision because the edge acts as an infinite boundary, effectively maximizing W .

The Human Information Processing (HIP) Model

- HCI researchers view the human mind similarly to a computer system. The HIP model divides human cognition into three interacting subsystems:
 1. **The Perceptual System:** Takes in data from the outside world (eyes, ears, touch). It is extremely fast but can easily be overwhelmed by too much stimuli (like a website with 50 flashing ads).
 2. **The Cognitive System:** The “CPU” of the brain. It processes the information, accesses memory, and makes decisions.
 3. **The Motor System:** Executes the physical response based on the cognitive system’s decision (e.g., moving the hand to click the mouse).

Human Memory and Its Limits

- Understanding the severe limitations of human memory is the most critical part of software design.
- **Short-Term / Working Memory (The Brain's RAM):** This holds information we are currently thinking about. *It is incredibly limited and volatile.*
 - ✓ **Miller's Law:** In 1956, George Miller discovered that the average human can only hold 7 ± 2 **items** in their working memory at one time.
 - ✓ **HCI Application (Chunking):** To bypass Miller's Law, we group information into "chunks."
- This is why phone numbers are written as +1 (800) 555-0199 instead of 18005550199.
- The hyphens and spaces do the cognitive work for the user.

- **Long-Term Memory (The Brain's Hard Drive):** Massive capacity, but retrieving information requires significant mental effort.
 - ✓ **HCI Application (Recognition vs. Recall):** It is much easier for humans to *recognize* something familiar than to *recall* it from memory. This is the entire reason Graphical User Interfaces (GUIs) were invented. It is easier to recognize and click a “Print” icon than it is to recall and type the command `lp -d printer_name file.txt` in a blank terminal.

Gestalt Principles

- How does the brain make sense of what the eyes see? The **Gestalt Principles** explain how humans naturally group visual elements together. Good UI designers use these principles to organize screens without needing extra dividing lines.
 - **Law of Proximity:** Objects placed close together are perceived as a related group. (e.g., A “Submit” button must be physically close to the form fields it applies to).
 - **Law of Similarity:** Objects that look alike are perceived as having the same function. (e.g., If one clickable link is blue, all clickable links must be blue).
 - **Law of Continuity:** The human eye naturally prefers to follow lines, curves, and sequences. Good UI design guides the eye smoothly across the screen without jarring interruptions.
 - **Color Vision Limitations:** About 8% of men are red-green colorblind.
- HCI Rule:* Never use color as the *only* way to convey information. A success message shouldn't just be a green box; it must be a green box *with a checkmark icon*.

1. Cognitive Ergonomics

- **The Goal** of cognitive ergonomics is to make the software completely intuitive. A well-designed cognitive interface feels almost invisible because it perfectly matches how the human brain naturally thinks and expects things to work.
- In **cognitive design**, we classify mental effort into three types of Cognitive Load:
 - a. **Intrinsic Cognitive Load**
 - This is the inherent difficulty of the task itself, independent of the interface.
 - ✓ *Example:* Calculating an insurance premium or parsing a complex database schema. The designer cannot reduce this load, but they must protect the user's mental space to handle it.

1. Cognitive Ergonomics

b. Extraneous Cognitive Load

- This is mental effort wasted on poorly designed interfaces. It represents structural flaws in the UI that force the brain to work harder just to read the screen.
 - ✓ *Example:* Unclear navigation, unorganized data, low-contrast fonts, or chaotic layouts.
 - ✓ *Technical Fix:* Applying **Gestalt Principles of Visual Perception** (Proximity, Similarity, Continuity, and Closure) to visually group related information, reducing visual search times.

c. Germane Cognitive Load

- This is beneficial mental effort. It is the brain power spent constructing mental schemas and processing information deeply to learn a system.
 - ✓ *Technical Fix:* Maintaining strict platform consistency (e.g., using standard design design tokens and predictable UI patterns) so users can transfer schemas from other apps instead of building new ones from scratch.

1. Cognitive Ergonomics

Cognitive Error Engineering

- Human error is rarely random; it is typically a systematic failure of the interface to safeguard human cognitive limits. In cognitive engineering, we categorize errors using James Reason's **GEMS** (Generic Error Modelling System):
- **Slips:** The user has the correct goal, but executes the physical action incorrectly.
 - *System Cause:* Buttons are too close together, or touch targets are too small.
 - *HCI Solution:* Increase target padding; implement undo functionality.
- **Mistakes:** The user executes the action perfectly, but their initial plan or understanding was completely wrong due to a flawed mental model.
 - *System Cause:* The interface misled the user about what an action would do (e.g., a "Delete" button that actually permanently purges a database without warning).
 - *HCI Solution:* Implement clear dialog confirmations, double-verification patterns, and unambiguous visual cues.

2. Physical Ergonomics

- **Physical ergonomics** deals with the tangible, anatomical, and biomechanical side of technology. It studies how the human body interacts with physical hardware and workspaces, focusing heavily on comfort, safety, and health.
- The physical environment in which the system is used may influence how well it is accepted and even the health and safety of its users. It should therefore be considered in all design.
- **Core Focus Areas:**
 - **Repetitive Strain Injuries (RSI):** Long hours of clicking a mouse or typing can cause chronic conditions like Carpal Tunnel Syndrome. Physical ergonomics designs alternative hardware (like vertical mice or split keyboards) to keep wrists in a natural position.
 - **Posture & Fatigue:** Adjusting screen heights to prevent neck strain (the "tech neck" phenomenon) and choosing chairs that support the spine.
 - **Mobile & Interaction Design:** It's not just about desks. On mobile devices, physical ergonomics maps out the "Thumb Zone"—the area of a smartphone screen that a user can easily reach with one thumb without stretching or dropping the phone.

2. Physical Ergonomics

The Medical Consequences of Poor Physical Design

When a computer interface or workstation is poorly designed, it causes micro-traumas to the body that accumulate over time.

- **Repetitive Strain Injury (RSI):** Damage to tendons, nerves, and muscles caused by repeated, unnatural motions. Typing on a flat, poorly positioned keyboard for 8 hours a day without wrist support is a primary cause.
- **Carpal Tunnel Syndrome (CTS):** A specific, severe type of RSI where the median nerve in the wrist is compressed by swollen ligaments, causing numbness, tingling, and severe pain in the hand.
- **Computer Vision Syndrome (CVS):** Also known as Digital Eye Strain. Symptoms include dry eyes, blurred vision, and tension headaches caused by staring at glowing screens, poor screen contrast, or heavy glare from overhead lights.
- **Musculoskeletal Disorders (MSDs):** Chronic back, shoulder, and neck pain resulting from hunching over laptops (a condition casually known as “Tech Neck”) or using non-adjustable chairs.

2. Physical Ergonomics

- **The Goal** of the physical ergonomics is to protect physical health. No matter how brilliant a piece of software is, if using it for an hour makes a user's wrists throb or eyes strain, the system has failed.

2. Physical Ergonomics

- There are five major senses: **sight, hearing, touch, taste** and **smell**. Of these, the first three are the most important to HCI. Taste and smell do not currently play a significant role in HCI, and it is not clear whether they could be exploited at all in general computer systems, although they could have a role to play in more specialized systems (smells to give warning of malfunction, for example) or in augmented reality systems. vision, hearing and touch are central.

2. Physical Ergonomics

- There are a number of effectors, including the limbs, fingers, eyes, head and vocal system. In the interaction with the computer, the fingers play the primary role, through typing or mouse control, with some use of voice, and eye, head and body position.
- Imagine using a personal computer (PC) with a mouse and a keyboard. The application you are using has a graphical interface, with menus, icons and windows. In your interaction with this system you receive information primarily by sight, from what appears on the screen.

2. Physical Ergonomics

Health issues

There are a number of factors that may affect the use of more general computers. Again these are factors in the physical environment that directly affect the quality of the interaction and the users.

- **Performance:** Users should be able to reach all controls comfortably and see all displays. Users should not be expected to stand for long periods and, if sitting, should be provided with back support. If a particular position for a part of the body is to be adopted for long periods (for example, in typing) support should be provided to allow rest.
- **Temperature:** Extremes of hot or cold will affect performance and, in excessive cases, health. Experimental studies show that performance deteriorates at high or low temperatures, with users being unable to concentrate efficiently.

2. Physical Ergonomics

- **Lighting:** The lighting level will again depend on the work environment. Adequate lighting should be provided to allow users to see the computer screen without discomfort or eyestrain. The light source should also be positioned to avoid glare affecting the display
- **Noise:** Excessive noise can be harmful to health, causing the user pain, and in acute cases, loss of hearing. Noise levels should be maintained at a comfortable level in the work environment. This does not necessarily mean no noise at all. Noise can be a stimulus to users and can provide needed confirmation of system activity.
- **Time:** The time users spend using the system should also be controlled. it has been suggested that excessive use of CRT displays can be harmful to users, particularly pregnant women.

2. Physical Ergonomics

- **The use of color:** Colors used in the display should be as distinct as possible and the distinction should not be affected by changes in contrast. Blue should not be used to display critical information. If color is used as an indicator it should not be the only cue: additional coding information should be included. The colors used should also correspond to common conventions and user expectations. Red, green and yellow are colors frequently associated with stop, go and standby respectively. Therefore, red may be used to indicate emergency and alarms; green, normal activity; and yellow, standby and auxiliary function. These conventions should not be violated without very good cause.

Comparison of Cognitive and Physical Ergonomics

Feature	Cognitive Ergonomics	Physical Ergonomics
Primary Target	The Human Mind (Brain)	The Human Anatomy (Body)
Deals With...	Attention, memory, logic, and error prevention.	Posture, muscle fatigue, reach, and hardware layout.
HCI Example	Structuring a checkout page so the user doesn't get confused.	Designing a mouse that keeps the wrist at a natural, un-strained angle.
Worst-Case Failure	User frustration, confusion, and catastrophic user errors.	Physical pain, fatigue, and long-term joint or muscle injuries.

- Ultimately, great HCI requires a balance of both. Physical ergonomics ensures the user can comfortably sit at the computer or hold the device, while cognitive ergonomics ensures that once they look at the screen, they actually know what to do.

QUESTIONS?